

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			flame test (1) brick-red colour (1)	2			2		2
	(c)	(i)		$\text{Ba}^{2+}(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) \rightarrow \text{BaCO}_3(\text{s})$		1		1		
		(ii)		to ensure that all the carbonate ions were precipitated	1			1		1
	(d)			some of the carbonates decompose / evolve CO_2 on heating (1) this will result in a lower mass being recorded (1)		1	1	2		2
	(e)	(i)		calcium chloride contains a lattice of oppositely charged ions (that are attracted to each other) / in the solid state the ions cannot move freely (1) in the molten state / in solution the ions are free to move and carry electricity (1)	2			2		
		(ii)		in iodine pairs of atoms are bonded covalently to form molecules which are held together by van der Waals forces (1) these forces are weak / easy to break (1)	2			2		
		(iii)		bonding in magnesium consists of a regular array of metal ions surrounded by a 'sea' of delocalised valence electrons (1) when a force is applied the layers of metal ions slide over each other forming a new shape (1)	2			2		
				Question 9 total	9	6	3	18	0	11